

IMP DevOps Interview Questions

1 AWS & Cloud

1. Consider a VPC with only one subnet. There are no route tables, no security groups, no configurations — it's completely empty. Do you think it is a private subnet or a public subnet?
2. What is EBS and EFS? Explain clearly.
3. How do you set up a highly available application?
4. How many accounts have you managed in your project, and how do you configure them with Jenkins?
5. If you are not able to access the EC2 instance, what might be the issues?
6. What is your project architecture?
7. How many AWS accounts have you managed in your project?
8. How many types of S3 buckets are there?
9. What are Security Groups and NACLs, and what are the differences between them?
10. How do you reduce AWS costs in your project?
11. What is VPC peering and Transit Gateway, and how do you set them up?
12. Have you used load balancers? What kinds of load balancers have you used, and why are they important?
13. How do you set up a highly available application?
14. What is a NAT Gateway, and how does it work?
15. If an EC2 instance has 50GB storage, how do you attach extra storage to an existing instance?
16. What S3 buckets have you used?
17. Have you worked with AWS Lambda functions? How did you use them in your projects?

18. How many organizations are using AWS Lambda instead of EC2, and in which scenarios would you prefer Lambda over EC2?
 19. How do you design AWS platforms for production workloads?
 20. How do you handle AWS account and environment separation?
 21. How do you design a multi-tier network where the database has zero internet access?
 22. ALB vs NLB — when do you use each?
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2 Jenkins & CI/CD

1. What are parameters in Jenkins?
2. What are shared libraries and why is it used?
3. Jenkins architecture
4. How to configure cluster in Jenkins
5. How do secure jenkins pipeline while running
6. If you got issue when deploy stage what steps do you follow
7. What branching strategies are you using?
8. How to reusable the jenkins pipeline
9. How do you trigger a pipeline, how many types we have.
10. How many pipelines we have in Jenkins.
11. How many CICD stages will you follow in your current project?
12. Where do you store documentation from your current project?
13. How do you share jenkins file with your colleagues
14. How do you improve the jenkins cicd process

15. If nodes are not active or not working how do fix it, for example master node and slave node.
 16. Write a parallel Jenkins pipeline (Jenkinsfile)
 17. How to deploy a .NET application using a CI/CD pipeline
 18. Common issues faced during CI/CD pipelines and how to resolve them
 19. Debugging steps when an application is not accessible from the web
 20. What are the different ways to run and build pipelines? Explain Poll SCM and Webhooks.
 21. How do you run multiple jobs in Jenkins?
 22. If your Jenkins pipeline failed because the artifact is not accessible due to credential issues. What could be the possible reasons?
 23. If you launched the service and installed Jenkins, but Jenkins is not accessible, what might be the reason?
 24. How do you install plugins, and what are the steps to install plugins?
 25. What are pipeline stages, and can you write them?
 26. How do you migrate Jenkins from one server to another?
 27. How do you store credentials in Jenkins?
 28. For example, Docker credentials.
 29. How do you store username and password securely?
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3 Docker

1. How do you reduce Docker images?
2. What is docker volumes
3. Explain how to write a docker image for nodejs
4. How do you go into the container, what is the command to use?

5. What is docker architecture
6. What is cmd and entrypoint
7. How do you pass the port in Docker when writing a Dockerfile?
8. What is CMD and ENTRYPOINT in Docker?
9. Docker image optimization & reduce strategies
10. How to reduce Docker image size
11. Docker volumes and their use cases
12. Difference between COPY and ADD instructions
13. How to share Dockerfiles with team members
14. How to check running vs stopped containers
15. What Docker steps do you follow to create an image?
16. Just explain Dockerfile creation.
17. What is Docker Compose?
18. How do you reduce or reuse Docker image size?
19. Write a Docker Compose file.

4 Kubernetes

1. I have 5 pods — 3 are running and 2 are not running. What might be the issue?
2. Pod vs ReplicaSet vs Deployment — which one should you choose for application deployment?
3. Cluster security.
4. If one service is consuming too much CPU and memory, how do you check which component is causing high usage?

5. What is configmap and secrets
6. What is statefulset and daemonset
7. How do you scale up min and max replicas
8. High availability for eks cluster
9. How to configure monitoring to eks cluster.
10. Difference between Kubernetes and Docker Swarm
11. What is ReplicaSet
12. Writing a basic Pod manifest file
13. Static Pods explanation
14. HPA vs VPA (Horizontal vs Vertical Pod Autoscaling)
15. Node troubleshooting (checking active/inactive nodes)
16. High availability architecture for applications
17. What happens after running Node?
18. Write a Deployment manifest file?
19. What happens internally when you apply a Kubernetes manifest file?
20. What is taint and toleration in Kubernetes?
21. What load balancers are you using in Kubernetes, and why are you using them in your project?
22. How many Kubernetes service types are there, and how do you use them?
23. Why do we use an Ingress service in Kubernetes?
24. What is the default namespace in Kubernetes?
25. How do two pods or two namespaces communicate?
26. How do you troubleshoot and fix a CrashLoopBackOff error?

27. How do you design a highly available production application using Kubernetes?
 28. Can you write a Kubernetes manifest file for a Horizontal Pod Autoscaler (HPA)?
 29. How do you design a secure Amazon EKS cluster architecture?
 30. What is the node scale-up and scale-down time? How do you define minimum and maximum nodes?
 31. How do you secure workloads in EKS?
 32. Explain IRSA and why it's important.
 33. How do you expose applications in EKS?
 34. Cluster Autoscaler vs Karpenter?
 35. Can you write a script to detect NotReady nodes in an EKS cluster with proper error handling?
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5 Terraform

1. When creating EC2, S3, Lambda — how are resources created? Parallel or sequential? How does Terraform create resources?
2. How do you create 10 resources one by one?
3. How do you secure the Terraform state file?
4. Terraform lifecycle, terraform refresh, terraform import
5. Writing a VPC creation Terraform script
6. Recommended Terraform project folder structure
7. terraform destroy vs terraform taint
8. Securing Terraform state files
9. What is Terraform drift?
10. If two people are working on the same Terraform module, how does it work?

11. How do you remove a Terraform lock file so the next person can apply?
12. If someone created a resource from the AWS console, how do you sync it with the current Terraform state?
13. When you change a Terraform module (e.g., S3), what happens during the terraform plan?
14. How do you create 50 instances using Terraform?
15. What is “dynamic” in Terraform and where is it used in real projects?
16. If resources are created via AWS Console/CLI, how do you bring them under Terraform without recreating them?
17. How do you securely store DB usernames and passwords using Terraform?
18. How do you create and consume AWS Secrets Manager in CI/CD pipelines?
19. In production, when EBS storage is full, how do you extend storage without downtime?
20. How do you manage Terraform state in production?
21. How do you structure Terraform modules?
22. Have you used Terragrunt or Terraform testing?
23. How do you manage Terraform state locking and safely run Terraform in CI/CD pipelines?
24. How do you design reusable Terraform modules while supporting different environments?
25. How did you use Terraform workspaces in your project? Can you explain the folder structure?
26. Can you write a Terraform module for creating a VPC?

Git

1. git merge
2. git pull
3. git fetch

4. git conflict
 5. What are the commands called git cherry-pick, git config, and git rebase?
 6. PR and MR requests — how do they work?
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7 Monitoring & Observability

1. What are the activities from monitoring
 2. Do you know how to configure monitoring to eks cluster.
 3. How monitoring tools collect metrics
 4. Real-world monitoring strategies
 5. What observability stack have you used?
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8 Linux / Scripting / Automation

1. Are you able to write shell scripts, if I give a scenario.
 2. Do you have exp in python?
 3. Crontab usage with scheduling scenarios
 4. Write a Bash script to create and write log files in the current directory.
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9 Security & DevSecOps

1. How do you secure the Terraform state file?
2. Cluster security.

3. How do you manage secrets?
 4. What security checks do you add in pipelines?
 5. How do you design IAM policies?
 6. AWS IAM policies & writing a sample IAM JSON policy
 7. How do you store passwords in terraform, what is the best practice you followed secure the passwords.
 8. If your colleague tries to upload your project's important files to their personal account, how would you secure against this type of threat?
 9. What is Policy as Code?
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10 Project / Process / Agile

1. What is agile methodology, and what about sprints release.
 2. How jira was used, and explain to me.
 3. Explain your project architecture
 4. CI/CD flow followed in your project
 5. Branching strategy (Dev, QA, Staging, Prod)
 6. How do you manage promotions across environments?
 7. GitOps with Argo CD — how does it work?
 8. What advanced Argo CD patterns have you used?
 9. Discuss how AI agents are important in modern technology.
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Ansible

1. What is Ansible and how does it work?
2. What is idempotency?
3. Difference between playbook and ad-hoc commands.
4. Difference between command and shell module.
5. What are roles in Ansible?
6. What is an Ansible Vault?
7. What is inventory?